EXPERIMENT NO.: 13

TITLE: CASE STUDY ON LIMITATIONS OF LOW-CODE APPLICATION DEVELOPMENT

***Roll Number:** 241401087*
***Platform:** Zoho Creator*
***Application:** Loan Application and Pre-screening Management*

1. Aim

To study the limitations of low-code application development by examining a practical application, identifying technical and operational challenges, and understanding when custom programming or external services may be required.

2. Software Requirements

- Zoho Creator account*
- Web browser and internet connection*
- Loan Application and Pre-screening Management application*

3. Theory

Low-code development is an approach to software development that uses visual interfaces, drag-and-drop components, predefined templates, workflows, and limited coding. It allows applications to be developed more quickly than traditional programming for many common business requirements.

However, low-code platforms have limitations when applications require complex algorithms, specialized integrations, detailed control over performance, or highly customized user interfaces.

Major limitations

***1. Limited customization:** Some platform components have predefined behavior and may not support every specialized requirement.*

***2. Platform dependency:** Applications may depend on the platform's features, pricing, availability, and supported technologies.*

***3. Complex integrations:** Integrating with specialized third-party systems may require APIs, custom functions, or external services.*

***4. Performance constraints:** Large datasets and complicated workflows may require careful optimization or a different architecture.*

***5. Security and governance:** Access permissions, data sharing, validation, and privacy must be configured correctly.*

***6. Scalability:** Applications with rapidly increasing traffic or complex processing requirements may need additional services or architectural changes.*

***7. Vendor lock-in:** Moving an application to another platform can require redesigning forms, workflows, integrations, and data structures.*

4. Case Study: Loan Application and Pre-screening Management

The application manages loan applicant information, employment details, income, loan requirements, and existing liabilities. Zoho Creator provides forms, reports, subforms, and workflow capabilities that help implement these requirements with limited coding.

Limitations identified

*****A. Complex financial calculations*****

Loan eligibility may depend on income, monthly liabilities, interest rates, loan duration, and institution-specific policies. Advanced calculations may require custom functions and extensive testing.

*****B. External credit verification*****

Connecting to an external credit bureau may require authentication, API integration, consent management, and secure handling of responses.

*****C. Advanced risk analysis*****

Sophisticated fraud detection or predictive risk scoring may require external analytical services or custom-developed models.

*****D. Large-scale processing*****

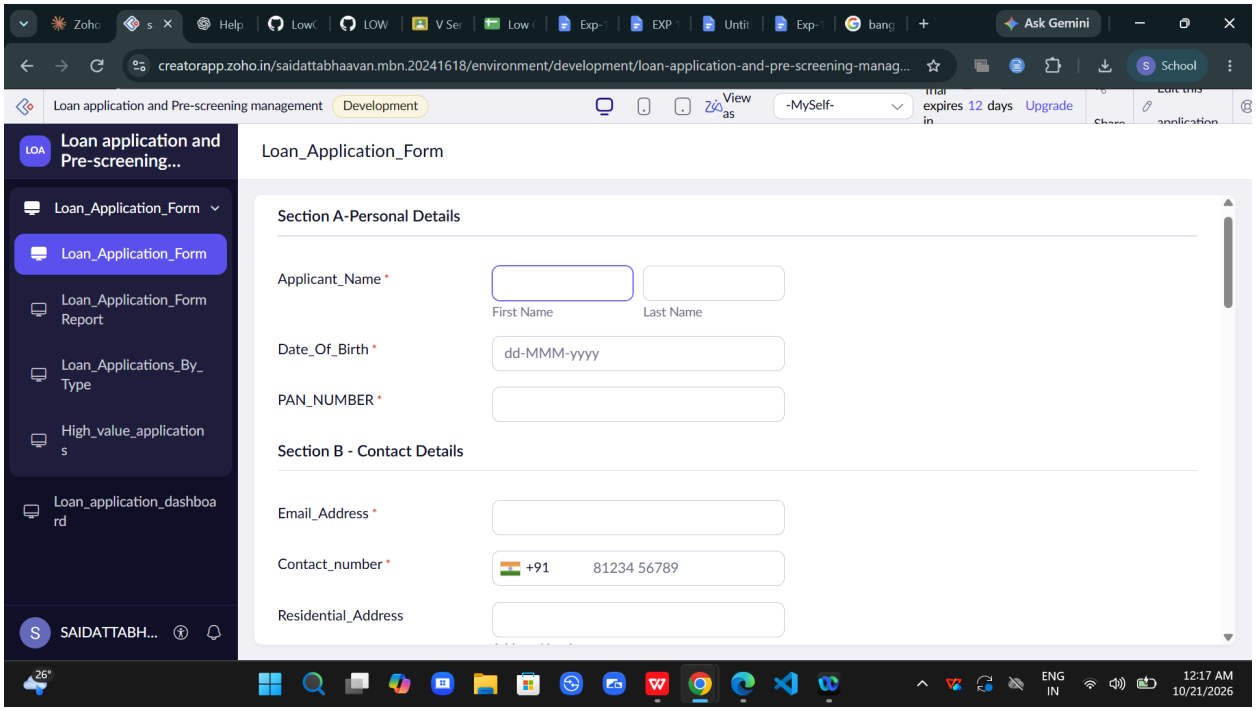
As the number of applications increases, reports and workflows may need optimization. Performance should be tested using representative data.

*****E. Data security*****

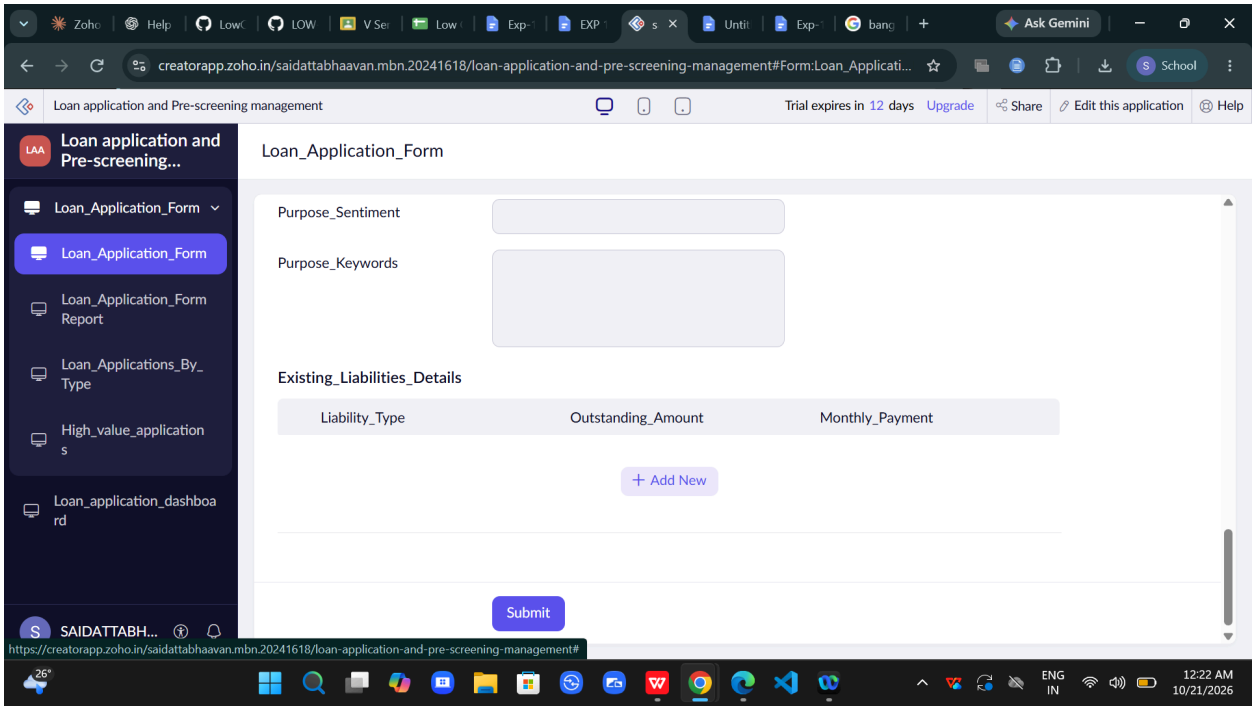
Financial and personal information requires appropriate access controls, secure integrations, careful data retention, and compliance with applicable requirements.

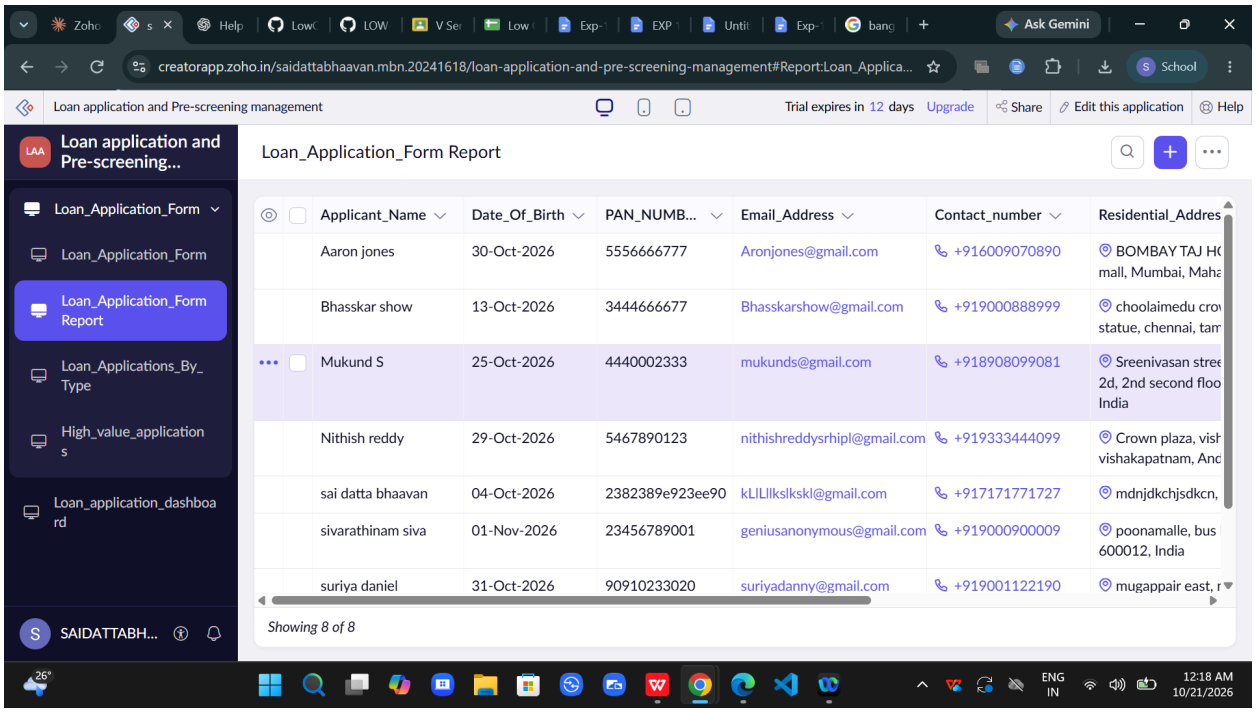
5. Procedure

1. Open the Loan Application and Pre-screening Management application in Zoho Creator.



2. Inspect the existing loan application form and identify its main fields.
3. Examine reports and subforms used to organize application records and existing liabilities.





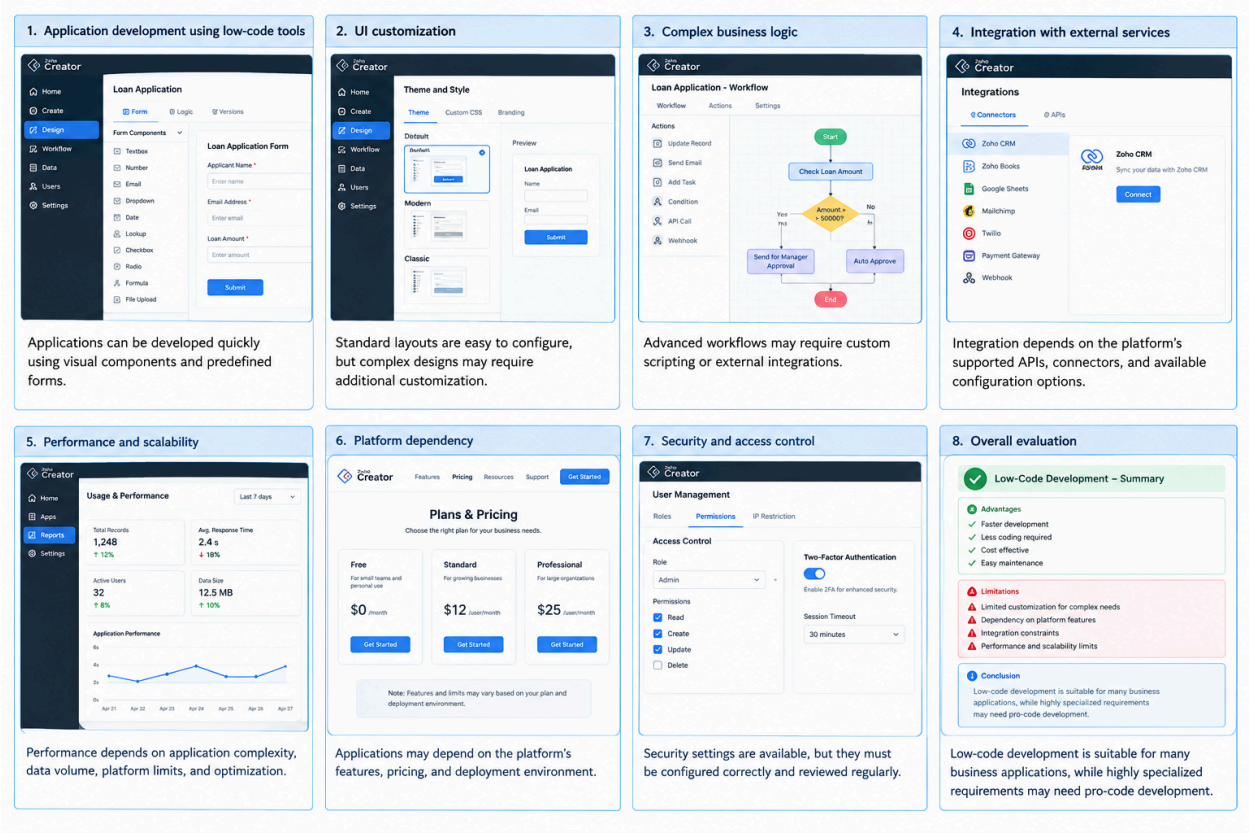
4. Identify requirements that can be implemented using built-in form, report, and workflow features.

5. Identify requirements that may need custom code, API integrations, or external services.

6. Compare the advantages and limitations of implementing these requirements using low-code tools.

7. Record the findings and suggest suitable improvements.

6. Observations



7. Suggested Improvements

1. Use custom functions for complex calculations.
2. Integrate external services through authenticated APIs where necessary.
3. Optimize reports, queries, and workflows as data volume increases.
4. Apply least-privilege access controls and protect confidential applicant data.
5. Test integrations, calculations, and failure scenarios before deployment.
6. Evaluate platform limitations before selecting a low-code solution for a critical business process.

8. Result

The limitations of low-code application development were studied using the Loan Application and Pre-screening Management application as a case study. The analysis identified requirements that can be handled by built-in platform features and those that may require custom programming, external integrations, or additional performance and security measures.

9. Conclusion

Low-code platforms simplify application development and reduce development time, but they do not eliminate the need for technical design, testing, security, and maintenance. A hybrid approach combining low-code components with custom programming and external services can address more complex requirements when necessary.